Coupling the Point Model to the Growth Field

Key points of the meeting of 20 October 2026

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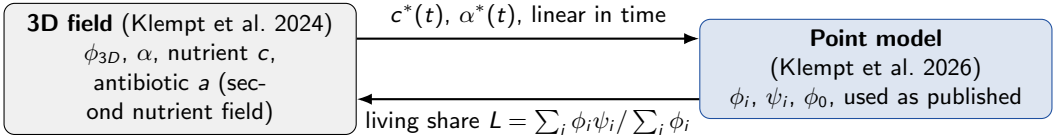
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What changes

- The point model of Klempt et al. 2026 is used **as published**: equations, parameters and its constraint $\sum_i \phi_i + \phi_0 = 1$ unchanged. It is called at every Gauss point with given inputs and returns its state.
- No longer: scaling the point model to the field amount ($\phi_0 = 1 - \phi_{3D}$).
- Field to point model: nutrient c^* and antibiotic α^* from 3D fields, interpolated in time.
- Point model to field: only the living biomass consumes nutrient and takes up antibiotic.
- Growth law Eq. 36 unchanged; Eq. 34 with its front term, so the nutrient reaches growth and stress.
- Into the thesis (Chapter 4), replacing the earlier coupling.

The loop at one Gauss point



L scales the consumption of c and a $\rightarrow$ c enters the front term of Eq. 34 $\rightarrow$ spreading of ϕ_{3D} $\rightarrow$ growth α (Eq. 36) $\rightarrow$ stress

Equations

inputs, $t_n \le t \le t_{n+1}$: $\quad c^*(t) = c_0^* \left[\frac{t_{n+1}-t}{\Delta t} c_n + \frac{t-t_n}{\Delta t} c_{n+1} \right], \quad \alpha^*(t) = \alpha_0^*\, a(t)$ likewise

nutrient (Eq. 35): $\quad 0 = d\,\nabla^2 c - g\,\phi_{3D}\,L\,c$

antibiotic: $\quad 0 = d_a \nabla^2 a - g_a\,\phi_{3D}\,L\,a$

spreading (Eq. 34): $\quad \dot{\phi} = \beta \nabla^2 \phi + k_\alpha \alpha - \|\nabla \phi\| \frac{r\,c}{k+c}\, \mathbf{n}_{\nabla\phi} \cdot \mathbf{n}_{\nabla c}$

growth (Eq. 36): $\quad \dot{\alpha} = k_\alpha\, \phi$

- $L = 1$ gives back Eq. 35; a dead biofilm ($L = 0$) consumes nothing. ϕ_0 has no living fraction.
- Eq. 35 is printed as $g\phi$; the implementation uses $g\phi c$ (misprint, confirmed).
- The antibiotic acts on the stress only indirectly: $\psi_i \downarrow \to L \downarrow \to c \uparrow \to$ front term.

Time: one clock for both models

- Klempt et al. 2024 normalise time with a reference time chosen for the problem. I choose $T^* = 1$ as the duration of the point model's data.
- The point model then advances $s\,\Delta T^*$ per increment, with s the length of that window in its own units: s **follows, it is not fitted.**
- Synthetic data of Fritsch et al. 2025, case I: 1000 steps of 10^{-4}, so $s = 0.1$.
- Checked with synthetic data (four species): with its own constraint and s from the window, the coupled point model reproduces the point model alone to 7×10^{-7}. Scaled to the field amount instead, the error is 0.019, and a fitted s comes out a third too low.

Examples and checks

Examples (synthetic, Fritsch et al. 2025 case I, two species, $b = (1, 2)$)

- A in the centre seed, B in the four vertical edges
- A and B mixed in the seed from the start
- nutrient from one wall, antibiotic from the opposite wall from $T^* = 0.5$
- each with and without antibiotic

Checks

- uniform c and a: the coupled point model equals the point model alone
- ANSYS against Python step by step (same inputs)
- ϕ stays in $[0, 1]$ with the front term
- sensitivity: $\phi_{\min}$, coupling step, mesh

Python runs in progress; ANSYS on 21 October.

Assumptions and open points

- Assumptions (not in the papers): linear interpolation of c^*, α^* in time; L from the previous increment; start values of newly reached points (mean state of the points the biofilm grows from); threshold $\phi_{\min} = 0.01$.
- The point model has no transport between points: species A and B meet only through the start values of new points.
- The front term as printed moves the biofilm towards the nutrient; its large cell Péclet number needs upwinding (overshoot check).
- The calibrated five-species model has $b = 0$ (no antibiotic in the data): the antibiotic is shown with synthetic data only.

Appendix: notation

ϕ_{3D}, α	biofilm phase field and growth ($\mathbf{F}_g = \alpha\mathbf{I}$, $\alpha(0) = 1$), Klempt et al. 2024
c, a	nutrient and antibiotic fields (the element's first and second nutrient)
d, g, d_a, g_a	diffusion and consumption of nutrient and antibiotic
β, k_α, r, k	diffusion of ϕ, growth rate, front growth and Monod constant (Eq. 34, 36)
ϕ_i, ψ_i, ϕ_0	volume fraction and living fraction of species i, auxiliary void (Klempt et al. 2026)
c^*, α^*, b_i	nutrient and antibiotic of the point model, antibiotic sensitivity
L	living share of the point model's biomass, $\sum_i \phi_i\psi_i / \sum_i \phi_i$
T^*, s	normalised time; point-model time per unit T^*